

Spring Data JPA Links (Bookmark them!)

1. API Docs : <https://docs.spring.io/spring-data/jpa/docs/2.7.18/api/>

2. Ref docs : <https://docs.spring.io/spring-data/jpa/docs/2.7.18/reference/html/>

Latest Spring java docs

Link : <https://docs.spring.io/spring-framework/docs/5.3.26/javadoc-api/>

Today's Topics

1. Revise -

RESTful web services overview.

RESTful web service related annotations

@ResponseBody

@RestController

@RequestBody

@PathVariable

@CrossOrigin

2. Exam objective

Content negotiation

Refer - "day14_help\Content Negotiation.txt"

3. Using swagger

Refer - "day14-data\day14_help\Regarding Swagger.txt"

Test the application with swagger as a REST client.

Enter Advanced REST

2 E-R with REST API

Un comment the association from Category class.

Generate suitable constructor

Exclude association fields from toString , to avoid recursion n thus leading to stack overflow

2.1 Run the end point - get all categories

Observation - Exception

Details - org.springframework.http.converter.HttpMessageNotWritableException: Could not write JSON: failed to lazily initialize a collection of role: com.blogs.entities.Category.posts, could not initialize proxy - no Session

Cause -Category 1->* Post

Default fetch type : LAZY (i.e post details are NOT lifted from DB -- represented by a proxy) --can't be access outside the session scope.

Soln -- change LAZY --> EAGER

2.2 Eager fetching ?

Observation - stack overflow to due recursion

root cause - bi dir association

(during ser Jackson from category --tries to ser posts n from post tries to ser. Category)

(For understanding the problem - run postman client)

2.3 Use @JsonIgnore (jackson annotation) to tell Jackson(vendor for ser n de-ser) to skip a field during ser n de-ser.

Not an effective solution

Reason - If REST clnt needs to fetch category + post details - it won't be possible, since @JsonIgnore is used !!!

2.4 Clean n recommended solution -

DO NOT expose entities for data exchange between REST clnt n REST server , use them only for persistence.

Instead use DTO(Data transfer obj) pattern to exchange data between client n server.

1 Entity --> DTO (is required during ser.)

Create DTO instance , read data from entity(getter) --> set the data in DTO (setter)

DTO --> Entity (is required during de-ser.)

Create entity instance , read data from dto(getter) --> set the data in entity (setter)

2. Instead of doing it ourselves --there are ready made 3rd party libs - meant for src obj --> dest obj conversion.

eg : ModelMapper, MapStruct

Refer - "day14_help\Advanced REST\readmes\Regarding ModelMapper.txt"

Objective - Create CategoryDTO , with required fields.

Control property access (@JsonProperty) , if required.

Modify service layer , to return List of Category entity or DTO ?

2.1 Replace actual Resource(eg : Employee / Customer / User / Category) by ResponseEntity

Refer - "Regarding ResponseEntity.txt"

Solve

Add new end point to get category n all blog posts associated

URL - http://host:port/categories/{categoryId} /posts

URI variable - category id

Resp - category + post details(id , cr tm , update ts, title , desc , contents)

3. Next objective

Desc - Add new Blog Post

* URL - http://host:port/posts

* Method - POST

* payload - BlogPostDTO (category Id , author id , title content , desc)

* Successful Resp - SC 201 Category + mesg (ApiResponse)

* Error resp - SC 400 , error mesg -wrapped in DTO(ApiResponse)

Lab work for more practice (Project help)

1. Post a comment

Desc - Post a comment

URL - http://host:port/comments

Method - POST

payload - Comment request DTO - comment text , rating , commenter id , blog post id

Successful Resp - SC 201 , Comment Resp DTO - comment id , text , rating , created on , updated on

Error resp -SC 400 , ApiResponse with error message.

2. user signup

Desc - User Sign up

URL - http://host:port/users/signup

Method - POST

payload - User request DTO - firstname , lastname , email ,pwd, dob , reg amount , role

(Can we reuse earlier UserRespDTO here ? If yes HOW ?)

Successful Resp - SC 201 , User resp dto

Error resp -SC 400 , ApiResponse with error message.

3. Assign user address

i/p - user id n address details

resp - message within ApiResponse

Add service layer method

AddressService n it's imple class

ApiResponse assignUserAddress(Long userId, AddressDTO dto)

4. Get all user names , dob , email from specified city

i/p - city

o/p - selected user details from the specified city.

5. Update Comment text.

i/p - comment id n next text

o/p - comment resp dto

6. Link Adhaar Card n languages (revision of value types in JPA/Hibernate)

i/p - user id , Adhaar Card n languages

o/p mesg

7. Delete all the comments by given commenter

i/p - commenter id

o/p - mesg (indicating how many comments were deleted)

8. Add New Tag

i/p - tag dto (tag name)

o/p - tag resp dto (id , creation time , updation time, name)

9. Assign post to the tag

i/p - tag name n post title

o/p - mesg

10. Remove the post from a tag.

i/p - tag id n post id

o/p - mesg

4. Objective : User signin endpoint

URL : http://host:port/users/signin

Method : POST (for adding email n pwd : in request body as additional security.)

Desc - User signin

- * URL - http://host:port/users/signin
- * Method - POST
- * payload - dto (email n pwd)
- * Successful Resp - SC 200, user details - all (dto)
- * Error resp - SC 400 , error mesg -wrapped in DTO(ApiResponse)

Solve this.

4.1 Next , Answer this

What will happen if you dont' add email or password , for user sign in ?

Observe n conclude.

Since there is no Presentation logic validation , request is reaching the DB n selecte query is fired UN NECESSARILY!!!!

5. How will you add server side validations ?

Refer - "day14-data\day14_help\Advanced REST\Advanced REST\readmes\regarding spring boot exc handling n validations.txt"

Steps :

add dependency : spring boot validation stater
(already added in pom.xml)

1.1For validating Request Body -

eg : Appointment DTO , User request DTO

DTO : fields

@NotEmpty , @NotBlank , @NotNull , @Pattern, @Min,@Max, @Range, @Length.....

@RequestBody + @Valid : req body : method arg of POST/PUT/PATCH

In case of validation failures , SC throws -

MethodArgNotValidException

It sends status code - SC 400 => BAD REQUEST

For validating -

1.2 path variables

1.3 request params

@Validated : RestController class level annotation

@NotBlank , @NotNull , @Pattern, @Min,@Max, @Range, @Length....

on req params or path vars

Exception raised : ConstraintViolationException

1.4 Why do we need to add centralized(Global) exc handling ?

In absence of custom exc handler , Spring boot supplies with default exception handler

--> sends JSON error object : timestamp , error code , err mesg ,err stack trace to the clnt !!!

If the exc : B.L failures (service layer excs) --try- catch block --too many try catch blocks in Rest controller methods

Such a try-catch block CAN NOT handle : validation failures

Soln : add custom (global) exc handler

How will you add it ?

1. Create a separate global exc handler class -

Add class level anno

@ControllerAdvice | @RestControllerAdvice

=> Spring bean containing common exc handling advice .

This exc handler is supplying a common advice to all REST controllers --
not to add exc handling logic separately BUT add it ONLY in this class.

2. Add method level annotation per exc handling method/s

eg : ResourceNotFoundException

@ExceptionHandler(ResourceNotFoundException.class)

```
public ResponseEntity<?> handleResourceNotFoundException(ResourceNotFoundException e)
{
    return ResponseEntity.status(HttpStatus.NOT_FOUND).body(new ApiResponse(e.getMessage()));
}
```

6. Spring Boot AOP (exam objective)

Refer - "day14_help\AOP\readme_aop.txt"

Spring Boot AOP

Objective If you want to intercept all the requests sent to this application n maintain a server side log ,
how will you do it ?

Refer : "day14_help\AOP\readme_aop.txt"

Refer for Java Functionnal Streams programming

<https://stackify.com/streams-guide-java-8/>